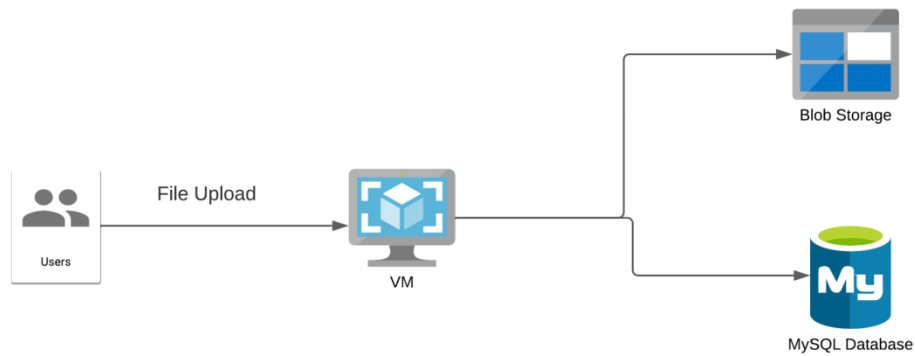


Declaration	
Questions in this exercise are intentionally complex and could be convoluted or confusing. This is by design and to simulate real-life situations where customers seldom give crystal clear requirements and ask unambiguous questions.	
I have read the above statement and agree to these conditions	
I AGREE	Esmeralda C. Cabrera Ventura
	<Enter your name above this line to indicate that you are in agreement>

Instructions
Every screenshot requested in this workbook is compulsory and carries 1 marks
Your Azure account ID must be clearly visible in every screenshot using the Azure portal; missing id or using someone else's id is not permitted. Such cases will be considered as plagiarism and severe penalty will be imposed.
All screenshots must be in the order mentioned under "Expected Screenshots" for every step
DO NOT WAIT UNTIL THE LAST MINUTE. The program office will not extend the project submission deadline under any circumstances.
The file should be renamed in the format BATCH_FIRSTNAME_LASTNAME_PROJECT1. For example: PGPCCMAY18_VIJAY_DWIVEDI_PROJECT1.pdf

Resource Clean Up
Cloud is always pay per use model and all resources/services that we consume are chargeable. Cleaning up when you've completed your lab or project is always necessary. This is true whether you're doing a lab or implementing a project at your workplace.
After completing the lab, make sure to delete each resource created in reverse chronological order.

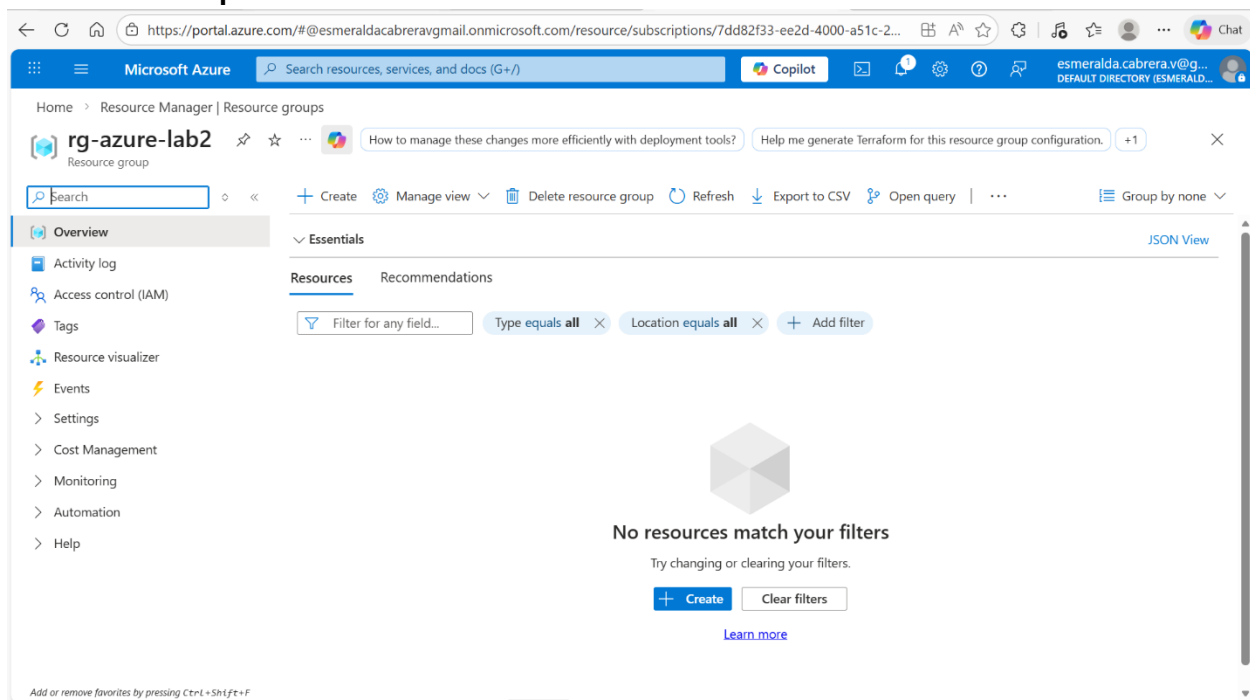
Architecture diagram**Architecture Implementation**

1	Upload the custom program and provided text file to a VM created using Ubuntu
2	Create a MySQL server using Azure Database service
3	Create a database inside the MySQL server created above
4	Running the custom program will convert the text file into a CSV file, upload it to blob storage and send the data to the MySQL server.

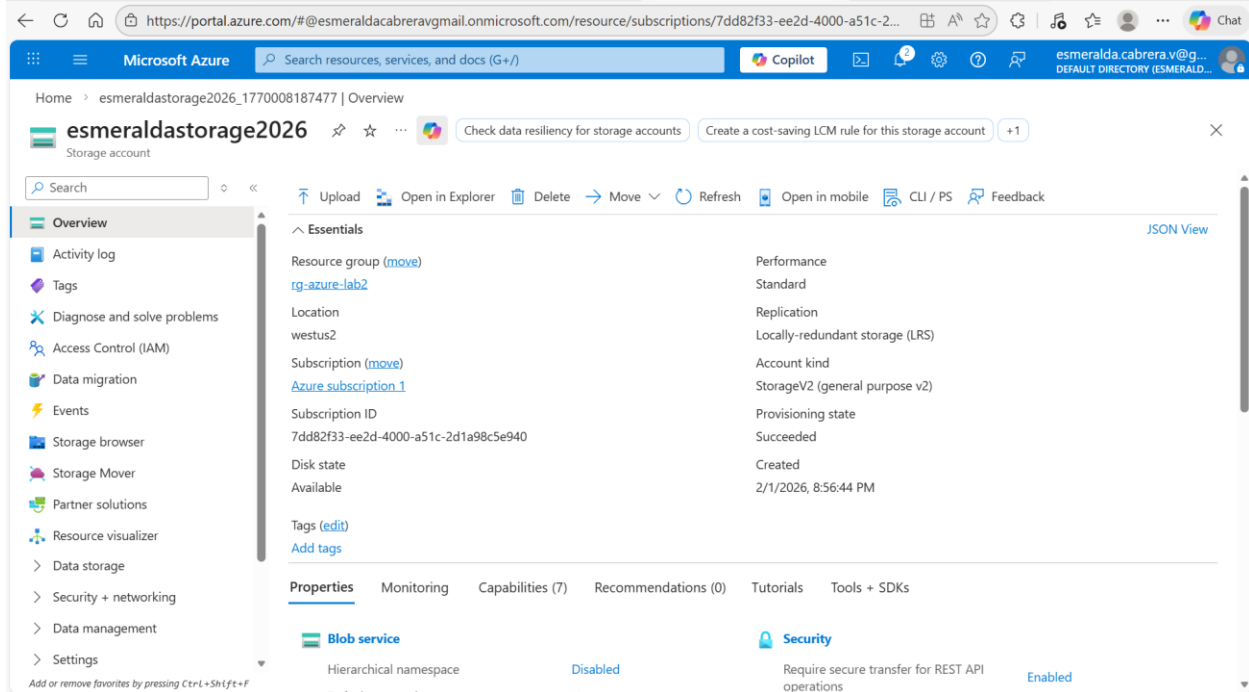
Step 1: Create resources

Step number	a
Step name	Creation of Resource group and blob storage
Instructions	1) Create a resource group using any region. Use the same resource group for all resources created in this exercise. 2) Navigate to Storage Accounts and Click on Create. 3) Enter a name and region for the Storage Account. The rest of the fields can be left to their default values. 4) Once the storage account has been created, navigate to the resource. 5) Using the menu on the left, navigate to Access Keys and note down the Connection String value for key 1. You may have to click on the Show keys button at the top of the screen to make the values visible.
Expected screenshots	1) Screen showing created storage account

Resource Group Creation

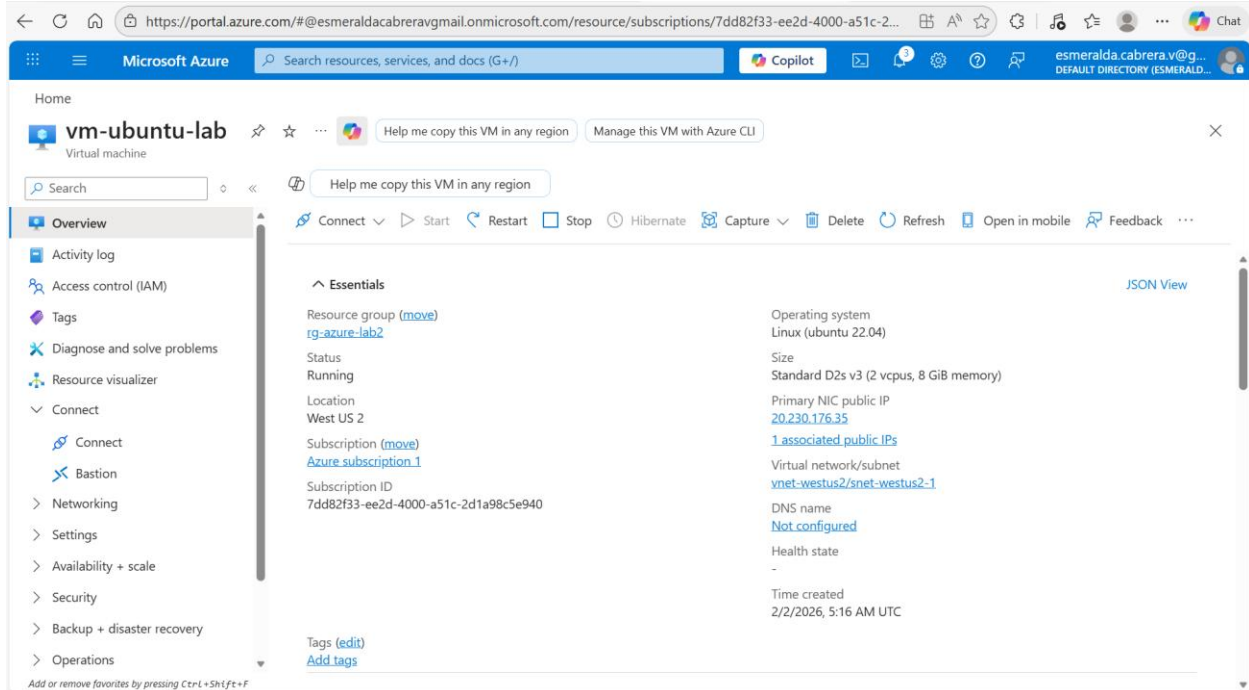


Storage Account Creation



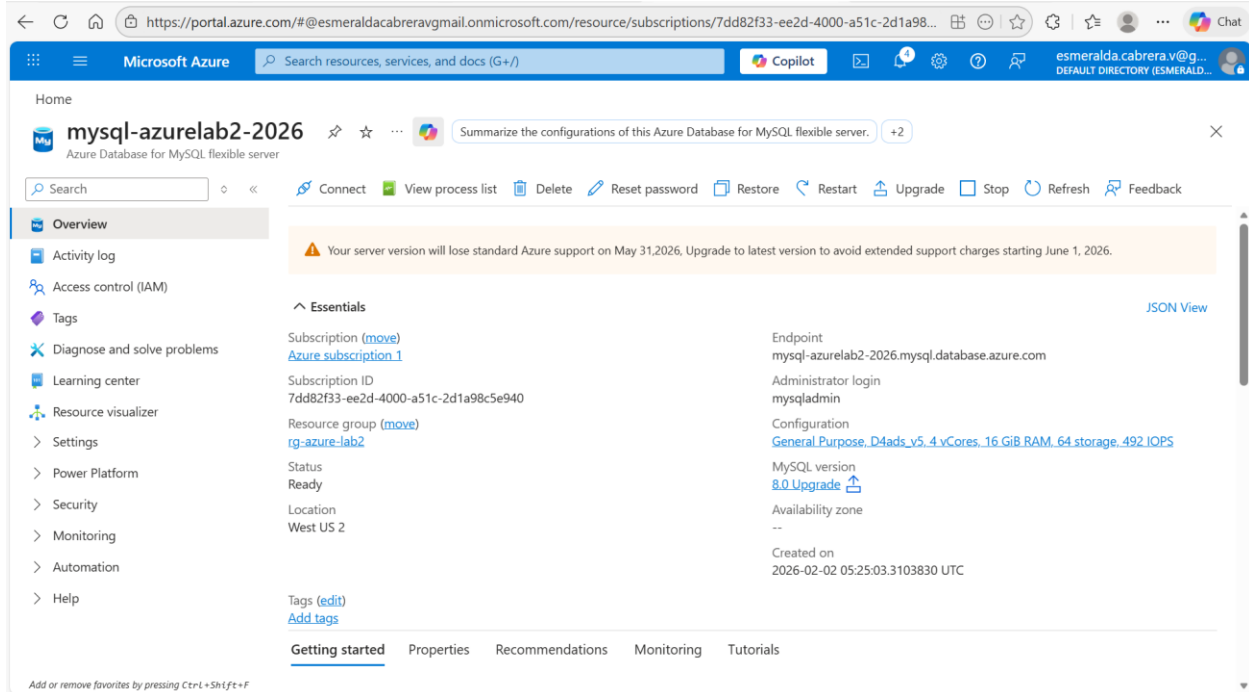
Step number	b
Step name	Creation of VM
Instructions	1) Navigate to Virtual Machines 2) Create a VM using the Ubuntu 20.04 image. 3) Make sure that port 22 is enabled in inbound ports for the VM during creation. 4) User name - ubuntu 5) Authentication type needs to be SSH public key. 6) The rest of the fields can be left to their default values. Click on Create.
Expected screenshots	1) Created VM

VM Creation



Step number	c
Step name	Creation of MySQL server
Instructions	1) Navigate to Azure Database for MySQL servers using the search bar at the top of the Azure portal and click on Create 2) Select the Flexible Server option 3) Enter the server name of choice and the username and password. Make sure to note down the username and password you have entered. 4) Under networking, ensure public access is allowed and check the box "Allow public access from any Azure service within Azure to this server" 5) The rest of the fields can be left to their default values. Click on Create. 6) Once the server has been created, navigate to the resource and note down the Server Name field present in the Overview section.
Expected screenshots	1) Overview screen of the created database server.

Creation of MySQL Database Server



Step 2: Run the custom program in the VM

Step number a

Step name Environment setup

- Instructions
- 1) Download the invoice file and python script provided with this workbook.
 - 2) Open the Python script using your text editor or code editor of choice
 - 3) Replace the values in lines 9,10,11, and 15 with the database server name, username, password, and storage account connection string(received in step 1(a)(5)) respectively. Save the file.
 - 4) Copy both the files to the VM using the scp command.
`scp -i <pem file> <file to be copied> ubuntu@<public IP of VM>:/home/ubuntu`
 You will need to run the scp command twice, once for each file.
 - 5) SSH into the VM using your SSH client of choice and run the below commands to set up the environment

```
sudo apt update
sudo apt install python3
sudo apt install python3-pip
sudo pip3 install pandas
sudo pip3 install azure-storage-blob
sudo pip3 install mysql-connector-python
sudo apt install mysql-client-core-8.0
```

- | | |
|----------------------|---|
| Expected screenshots | 1) Screenshot of the process.py file after completing Step3 above
2) Copying the files using scp
3) Screenshot after completing Step 5 above. |
|----------------------|---|

Process.py File after Completing Step 3 Above

```

process.py
File Edit View

import mysql.connector
import pandas as pd
import os, uuid
from azure.storage.blob import BlobServiceClient, BlobClient, ContainerClient, __version__
import csv

hostname = 'mysql-azurelab2-2026.mysql.database.azure.com'
username = 'mysqladmin'
password = '@ngelFire1'
database = "testdb"

def main():
    connect_str = 'DefaultEndpointsProtocol=https;AccountName=esmeraldastorage2026;AccountKey=AjB54/qoZsjvKvEu8MLL3DhXdnYfONWYVaFZ1k8Y2tpvfXUfWjX93Q/Q+3ixQz98SR93tqeyXce+ASTsXUezg==;EndpointSuffix=core.windows.net'
    # Create the BlobServiceClient object which will be used to create a container client
    blob_service_client = BlobServiceClient.from_connection_string(connect_str)
    # Create a unique name for the container
    container_name = str(uuid.uuid4())
    # Create the container
    container_client = blob_service_client.create_container(container_name)
    print('***** Processing File *****'),
    read_file = pd.read_csv(r'./docproc-invoice.txt')
    read_file.to_csv(r'./docproc.csv', index=None)
    print('File has been processed')
    upload_file_path = os.path.join("./", "docproc.csv")
    # Create a blob client using the local file name as the name for the blob
    blob_client = blob_service_client.get_blob_client(container=container_name, blob="docproc.csv")
    print("\nUploading to Azure Storage as blob: docproc.csv")
    # Upload the created file
    with open(upload_file_path, "rb") as data:
        blob_client.upload_blob(data)

data= pd.read_csv("./docproc.csv")
  
```

Copying the Process.py and Invoice Files using SCP Commands

```

ubuntu@vm-ubuntu-lab: ~
-a----- 11/3/2025 11:16 PM 59 costs.csv
-a----- 11/29/2025 11:27 PM 258 CSC670-65107-2512_report_Esmeralda_Cabrera_Ventura_71894_2025-11-29_2327_PST.csv
-a----- 11/27/2024 2:16 AM 123933 foodhub_order (1).csv
-a----- 12/8/2024 2:23 PM 123933 foodhub_order (2).csv
-a----- 11/25/2024 9:52 AM 123933 foodhub_order.csv
-a----- 6/30/2025 8:43 PM 1033 High-Protein_Meal_Plan.csv
-a----- 2/16/2025 6:30 PM 208964 marketing_campaign.csv
-a----- 6/3/2024 9:38 AM 169 payment-history.csv
-a----- 11/23/2024 1:42 AM 112852 StockData.csv
-a----- 11/29/2024 1:47 PM 848 Summary_Insights_from_Dataset.csv
-a----- 11/29/2024 3:10 PM 1487 Summary_of_Dataset_Insights.csv

PS C:\Users\Esméralda Cabrera> scp -i "C:\Users\Esméralda Cabrera\Downloads\vm-ubuntu-lab_key.pem" \
>> "C:\Users\Esméralda Cabrera\Downloads\azurelab2\script.py" \
>> ubuntu@20.230.176.35:/home/ubuntu/
C:\WINDOWS\System32\OpenSSH\scp.exe: stat local "C:\Users\Esméralda Cabrera\Downloads\script.py": No such file or directory
PS C:\Users\Esméralda Cabrera> dir "C:\Users\Esméralda Cabrera\Downloads\process*", "C:\Users\Esméralda Cabrera\Downloads\docproc-invo
ice*"

Directory: C:\Users\Esméralda Cabrera\Downloads

Mode                LastWriteTime         Length Name
----                -
-a-----         2/1/2026   9:39 PM           2492 process.py
-a-----         2/1/2026   9:33 PM           251 docproc-invoice.txt

PS C:\Users\Esméralda Cabrera> scp -i "C:\Users\Esméralda Cabrera\Downloads\vm-ubuntu-lab_key.pem" "C:\Users\Esméralda Cabrera\Downlo
ads\process.py" ubuntu@20.230.176.35:/home/ubuntu/
process.py                                     100% 2492    51.8KB/s   00:00
PS C:\Users\Esméralda Cabrera> scp -i "C:\Users\Esméralda Cabrera\Downloads\vm-ubuntu-lab_key.pem" "C:\Users\Esméralda Cabrera\Downlo
ads\docproc-invoice.txt" ubuntu@20.230.176.35:/home/ubuntu/
docproc-invoice.txt                           100% 251     4.6KB/s   00:00
PS C:\Users\Esméralda Cabrera> ls -l
  
```

SSH into Ubuntu Machine after Completing Step 5 Above

```

ubuntu@vm-ubuntu-lab: ~
PS C:\Users\Esmeralda Cabrera> ssh -i "C:\Users\Esmeralda Cabrera\Downloads\vm-ubuntu-lab_key.pem" ubuntu@20.230.176.35
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1044-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Feb  2 05:59:57 UTC 2026

System load:  0.03          Processes:           130
Usage of /:   6.3% of 28.89GB Users logged in:          0
Memory usage: 4%           IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Feb  2 05:46:56 2026 from 47.151.93.65
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@vm-ubuntu-lab:~$ ls -l
total 8
-rw-rw-r-- 1 ubuntu ubuntu 251 Feb  2 05:59 docproc-invoice.txt
-rw-rw-r-- 1 ubuntu ubuntu 2492 Feb  2 05:58 process.py
ubuntu@vm-ubuntu-lab:~$ sudo apt update
sudo apt install -y python3 python3-pip mysql-client-core-8.0
pip3 install --user pandas azure-storage-blob mysql-connector-python

```

All Installations Complete

```

ubuntu@vm-ubuntu-lab: ~
No VM guests are running outdated hypervisor (qemu) binaries on this host.
Collecting pandas
  Downloading pandas-2.3.3-cp310-cp310-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (12.8 MB)
    12.8/12.8 MB 48.5 MB/s eta 0:00:00
Collecting azure-storage-blob
  Downloading azure_storage_blob-12.28.0-py3-none-any.whl (431 kB)
    431.5/431.5 KB 37.4 MB/s eta 0:00:00
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.6.0-cp310-cp310-manylinux_2_28_x86_64.whl (34.5 MB)
    34.5/34.5 MB 40.2 MB/s eta 0:00:00
Requirement already satisfied: pytz>=2020.1 in /usr/lib/python3/dist-packages (from pandas) (2022.1)
Collecting tzdata>=2022.7
  Downloading tzdata-2025.3-py2.py3-none-any.whl (348 kB)
    348.5/348.5 KB 36.7 MB/s eta 0:00:00
Collecting numpy>=1.22.4
  Downloading numpy-2.2.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (16.8 MB)
    16.8/16.8 MB 72.6 MB/s eta 0:00:00
Collecting python-dateutil>=2.8.2
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
    229.9/229.9 KB 20.0 MB/s eta 0:00:00
Collecting azure-core>=1.30.0
  Downloading azure_core-1.38.0-py3-none-any.whl (217 kB)
    217.8/217.8 KB 23.8 MB/s eta 0:00:00
Collecting typing-extensions>=4.6.0
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
    44.6/44.6 KB 5.3 MB/s eta 0:00:00
Requirement already satisfied: cryptography>=2.1.4 in /usr/lib/python3/dist-packages (from azure-storage-blob) (3.4.8)
Collecting isodate>=0.6.1
  Downloading isodate-0.7.2-py3-none-any.whl (22 kB)
Requirement already satisfied: requests>=2.21.0 in /usr/lib/python3/dist-packages (from azure-core>=1.30.0->azure-storage-blob) (2.25.1)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Installing collected packages: tzdata, typing-extensions, python-dateutil, numpy, mysql-connector-python, isodate, pandas, azure-core, azure-storage-blob
WARNING: The scripts f2py and numpy-config are installed in '/home/ubuntu/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed azure-core-1.38.0 azure-storage-blob-12.28.0 isodate-0.7.2 mysql-connector-python-9.6.0 numpy-2.2.6 pandas-2.3.3 python-dateutil-2.9.0.post0 typing-extensions-4.15.0 tzdata-2025.3
ubuntu@vm-ubuntu-lab:~$ python3 --version
python3 --version
pip3 --version
pip3 --version
mysql --version
mysql --version

```


Step number	b
Step name	Configure the database
Instructions	1) Run the following command in the SSH terminal after substituting the database server name and username. <pre>mysql -h <database server name> -u <database_username> -p</pre> <i>Note : In case of a database server connection error , ensure that the firewall is configured correctly using the below link</i> https://learn.microsoft.com/en-us/azure/mysql/single-server/how-to-manage-firewall-using-portal 2) Enter the password when prompted. 3) Enter the following command <pre>create database testdb;</pre> 4) Enter <i>exit</i> to exit out of the MySQL environment.
Expected screenshots	1) Screenshot after completing Step 3 above

Configuring the database

```

ubuntu@vm-ubuntu-lab: ~
Downloading azure_core-1.38.0-py3-none-any.whl (217 kB)
217.8/217.8 KB 23.8 MB/s eta 0:00:00
Collecting typing-extensions>=4.6.0
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
44.6/44.6 KB 5.3 MB/s eta 0:00:00
Requirement already satisfied: cryptography>=2.1.4 in /usr/lib/python3/dist-packages (from azure-storage-blob) (3.4.8)
Collecting isodate>=0.6.1
Downloading isodate-0.7.2-py3-none-any.whl (22 kB)
Requirement already satisfied: requests>=2.21.0 in /usr/lib/python3/dist-packages (from azure-core>=1.38.0->azure-storage-blob) (2.25.1)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Installing collected packages: tzdata, typing-extensions, python-dateutil, numpy, mysql-connector-python, isodate, pandas, azure-core, azure-storage-blob
WARNING: The scripts f2py and numpy-config are installed in '/home/ubuntu/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed azure-core-1.38.0 azure-storage-blob-12.28.0 isodate-0.7.2 mysql-connector-python-9.6.0 numpy-2.2.6 pandas-2.3.3 python-dateutil-2.9.0.post0 typing-extensions-4.15.0 tzdata-2025.3
ubuntu@vm-ubuntu-lab:~$ python3 --version
Python 3.10.12
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)
mysql Ver 8.0.44-0ubuntu0.22.04.2 for Linux on x86_64 ((Ubuntu))
ubuntu@vm-ubuntu-lab:~$ mysql -h mysql-azurelab2-2026.mysql.database.azure.com -u mysqladmin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.42-azure Source distribution

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.02 sec)

mysql>

```

Step number	c
Step name	Running the custom program
Instructions	1) Run the program using the command <i>python3 process.py</i> 2) Navigate to the storage account using the Azure portal. Select the Containers option from the menu on the left and select the created container. Verify that it contains a generated CSV file 3) Run the following command in the SSH terminal after substituting the database server name and username. <i>mysql -h <database server name> -u <database_username> -p</i> 4) Enter the password when prompted. 5) Run the following commands to verify that the data has been entered into the database <i>use testdb;</i> <i>select * from invoice;</i> 6) Enter <i>exit</i> to exit out of the MySQL environment.
Expected screenshots	1) Running the custom Python program 2) Created CSV file in Blob Storage 3) Screenshot after running step 5 above

Running the .py program using the command `python3 process.py`

```

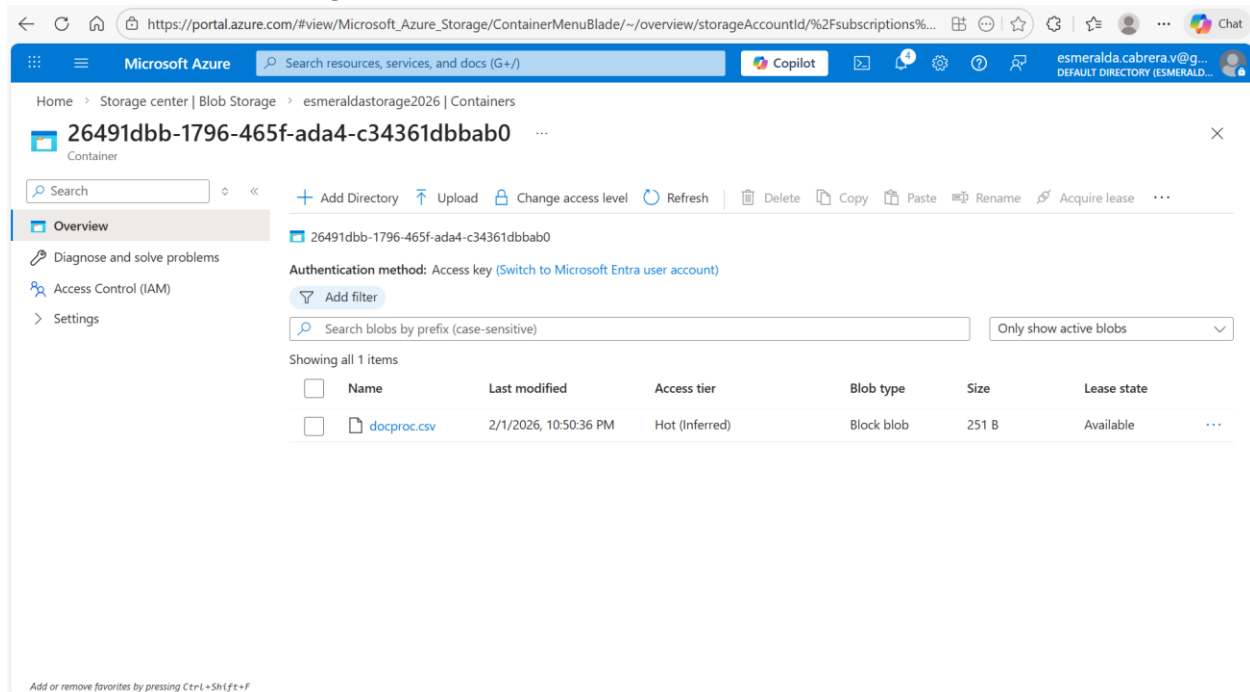
ubuntu@vm-ubuntu-lab: ~
raise error
azure.core.exceptions.ServiceRequestError: <urllib3.connection.HTTPSConnection object at 0x70ddf00b2350>: Failed to establish a new connection:
[Errno -2] Name or service not known
ubuntu@vm-ubuntu-lab:~$ scp -i "C:\Users\Esmeralda Cabrera\Downloads\vm-ubuntu-lab_key.pem" '
"C:\Users\Esmeralda Cabrera\Downloads\process.py" '
ubuntu@20.230.176.35:/home/ubuntu/
C:\Users\Esmeralda Cabrera\Downloads\process.py: command not found
usage: scp [-346ABCOpqRrsTv] [-c cipher] [-D sftp_server_path] [-F ssh_config]
          [-i identity_file] [-J destination] [-l limit]
          [-o ssh_option] [-P port] [-S program] source ... target
-bash: ubuntu@20.230.176.35:/home/ubuntu/: No such file or directory
ubuntu@vm-ubuntu-lab:~$ nano process.py
ubuntu@vm-ubuntu-lab:~$ nl -ba process.py | sed -n '7,20p'
 7
 8
 9 hostname = 'mysql-azurelab2-2026.mysql.database.azure.com'
10 username = 'mysqladmin'
11 password = '@ngelF!rel'
12 database = "testdb"
13
14 def main():
15     connect_str = 'DefaultEndpointsProtocol=https;AccountName=esmeraldastorage2026;AccountKey=AjB54/qoZsjvkEu8MLL3DhXdNYFONWyVaFZ1k8Y2tp
vfxuIFWjX930/Q+3ixsQz98SR93tqeyXce+AStSxUezg==;EndpointSuffix=core.windows.net'
16     # Create the BlobServiceClient object which will be used to create a container client
17     blob_service_client = BlobServiceClient.from_connection_string(connect_str)
18     # Create a unique name for the container
19     container_name = str(uuid.uuid4())
20     # Create the container
ubuntu@vm-ubuntu-lab:~$ python3 process.py
***** Processing File *****
File has been processed

Uploading to Azure Storage as blob: docproc.csv
inv-00001
Mar 31 2018

*****
Creating table invoice
Inserting data into database
ubuntu@vm-ubuntu-lab:~$

```

Creation of Blob Storage Container with CSV File



Home > Storage center | Blob Storage > esmeraldastorage2026 | Containers

26491dbb-1796-465f-ada4-c34361dbbab0 Container

Search resources, services, and docs (G+J) Copilot esmeralda.cabrera.v@g... DEFAULT DIRECTORY (ESMERALD...

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
docproc.csv	2/1/2026, 10:50:36 PM	Hot (Inferred)	Block blob	251 B	Available

Add or remove favorites by pressing Ctrl+Shift+F

Step 5: Data Entry Verification in Database

```
ubuntu@vm-ubuntu-lab: ~  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
New release '24.04.3 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Mon Feb 2 05:59:58 2026 from 47.151.93.65  
ubuntu@vm-ubuntu-lab:~$ mysql -h mysql-azurelab2-2026.mysql.database.azure.com -u mysqladmin -p  
Enter password:  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 16  
Server version: 8.0.42-azure Source distribution  
  
Copyright (c) 2000, 2025, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
mysql> USE testdb;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
mysql> SELECT * FROM invoice;  
+-----+-----+  
| cust_id | inv_id |  
+-----+-----+  
| inv-00001 | Mar 31 2018 |  
+-----+-----+  
1 row in set (0.00 sec)  
  
mysql>
```

Answer the following questions

Q1 At which level are lifecycle management rules for Blob storage applied?

- a) File Level
- b) Blob Level
- c) Storage account level
- d) Subscription level

Enter your answer here

Storage Account Level

Q2 Which of the following is **not** true about the Premium performance storage tier in Azure?

- a) Only Hot and Cool storage tiers are available
- b) Supports only LRS and ZRS
- c) Data is stored on SSDs
- d) Geo-redundancy is not possible.

Enter your answer here

**Only Hot and Cool Storage Tiers
Are Available**

Q3 Which of the following Azure SQL deployment options should you use when the number of databases to be created is variable.

- a) On-premises deployment of Azure SQL
- b) Azure SQL Database
- c) Managed DB instance
- d) None of these

Enter your answer here

Azure SQL Database

Q4 Which of the following Azure SQL purchasing models would be more beneficial for BYOL (Bring-Your-Own-License) use-cases?

- a) Depends on the license type
- b) Does not matter
- c) vCore based
- d) DTU based

Enter your answer here

vCore Based

Q5 Why was port 3306 not enabled for incoming connections in the VM in this exercise?

- a) The port is only required to be enabled on the database server
- b) Azure MySQL uses a different port
- c) Port 3306 has no bearing on this exercise.
- d) None of these

Enter your answer here

The Port Is Only Required to Be Enabled on the Database Server

Grades distribution

MCQs	10 (2 points each)
Implementation screenshots	10 points (1 point each)
Total	20 points